

Network Slice Embedding over Space Division Multiplexed Elastic Optical Networks

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Abstract—Network slicing over space-division-multiplexed elastic optical networks (SDM-EONs) enables efficient multi-service provisioning on a shared optical substrate. However, embedding such slices requires coordinated routing, modulation, core, and spectrum assignment (RMCSA) alongside compute resource management under dynamic traffic, which most existing RMCSA studies treat independently. This paper focuses on the network slice embedding problem over SDM-EONs, aiming to develop efficient resource allocation strategies that ensure both high utilization and reliable service performance. While prior studies have investigated RMCSA, they typically consider these dimensions separately from compute placement. To address this gap, this paper proposes a Waypoint-Assisted Multi-Segment Slice Mapping (WMSM) scheme, which integrates compute placement with spectrum allocation in a sequential but coupled manner to enable more flexible resource placement and improved spectrum efficiency. Numerical results show that WMSM improves acceptance ratios by up to 27% under high-load conditions, while achieving up to 47% lower total provisioning cost relative to the baseline strategy. These results highlight the benefits of integrated compute-spectrum provisioning and provide design insights for scalable, compute-aware optical slice mapping.

Index Terms—Network Virtualization, RMCSA, Spectrum Utilization, and Virtual Optical Network Embedding.

I. INTRODUCTION

The exponential growth of bandwidth-intensive applications continues to push the limits of optical transport infrastructures. Space-division multiplexing (SDM) combined with elastic optical networks (EONs) has emerged as a promising solution, enabling multiple spatial cores per fiber and dynamic spectrum allocation to deliver substantial improvements in capacity, flexibility, and spectral efficiency [1].

Network slicing, widely adopted in 5G to partition radio and core network resources into isolated virtual networks for diverse services [1], follows a similar principle in EONs but operates at the optical transport layer. Unlike 5G slicing, which primarily manages radio spectrum, computational, and backhaul resources through software-defined control, EON

slicing must additionally satisfy optical-layer constraints such as spectrum continuity, contiguity, and modulation reach. In SDM-EONs, slicing spans three dimensions: spectrum, spatial cores, and computational capacity. This introduces coupled constraints, that is, spectrum assignment must satisfy continuity and contiguity, and cores must be selected to mitigate inter-core crosstalk, and compute placement must respect node capacity limits, making joint optimization computationally intractable, while independent treatment leads to poor utilization and increased blocking.

Prior studies on network slicing over EONs typically treat compute placement separately from routing, modulation, core, and spectrum assignment (RMCSA), often leading to suboptimal or infeasible slice mappings: spectrum may be reserved along a path where compute capacity is unavailable, or vice versa. Most optical slice-embedding works also assume a strict one-to-one mapping between virtual and substrate nodes, which limits utilization. In the broader Virtual Network Embedding (VNE) literature, Wang *et al.* [2] and Chowdhury *et al.* [3] showed that node co-location and coordinated node-link mapping improve embedding efficiency and acceptance ratio, yet such strategies have not been extended to optical or SDM-EON slicing frameworks. Our work introduces controlled co-location of slice nodes on shared substrate nodes, improving compute utilization and reducing provisioning cost.

To address these gaps, this paper proposes two integrated heuristics for slice mapping in SDM-EONs. *Direct Path-Driven Slice Mapping* (DPSM) serves as a baseline, prioritizing direct end-to-end routing with sequential compute placement. *Waypoint-Assisted Multi-Segment Slice Mapping* (WMSM) introduces intermediate compute-capable waypoints and performs segment-wise spectrum allocation, enabling flexible, resource-efficient multi-segment provisioning. Both algorithms couple compute placement and spectrum allocation in a sequential but interdependent manner.

The main contributions are:

- A coordinated system model for SDM-EON slice mapping that integrates spectrum, compute, and inter-core crosstalk constraints in a sequential but interdependent manner.
- A WMSM algorithm evaluated against a DPSM baseline, both with and without balanced placement variants.
- A performance evaluation demonstrating significant reductions in blocking probability, spectrum usage, and provisioning cost over baseline prioritization schemes.

The rest of the paper is organized as follows: Section II reviews related work. Section III presents the system model and problem formulation. Section IV describes the proposed algorithms. Section V evaluates performance, and Section VI concludes.

II. BACKGROUND LITERATURE

In single-core EONs, several studies formulated the VNE problem in conjunction with Routing and Spectrum Allocation (RSA). Gong and Zhu [4] introduced one of the earliest Virtual Optical Network Embedding (VONE) formulations over EONs, proposing ILP and heuristic strategies based on layered-auxiliary-graph modeling. Wang and Hu [5] later improved scalability through a path-growing approach for optical VNE in SLICE networks. However, these works focused on spectrum utilization and connectivity without incorporating spatial multiplexing or compute-aware placement. Cross-haul and 5G transport studies [6], [7] integrated slicing across RAN and metro layers using heuristic-assisted deep reinforcement learning, but operate at higher architectural layers unsuitable for core-level optical provisioning.

With the emergence of multi-core fibers, research on SDM-EONs has addressed RCSA and RMCSA problems incorporating IC-XT constraints. Zhang *et al.* [8] and Kumar *et al.* [9] proposed crosstalk-aware VONE algorithms mitigating IC-XT in heterogeneous and spectrally-spatially elastic optical networks. Jin *et al.* [10] introduced a dynamic topology-aware VONE for hybrid SDM-EON services. Tang *et al.* [11] proposed an ILP model minimizing IC-XT jointly in spatial, frequency, and time domains. More recent heuristics [12], [13] introduced congestion-aware and group-processing algorithms for dynamic provisioning and fair spectrum allocation. While these studies address crosstalk, congestion, and fairness, they treat compute placement as independent of optical resource allocation.

Most optical VONE studies adopt one-to-one node mapping to avoid compute contention and single-point failures. In contrast, the broader VNE community has examined node co-location to improve embedding efficiency. Wang *et al.* [2] proposed *Node-Fusion*, evaluating substrate nodes via an integrated ranking metric based on CPU availability, node degree, and link bandwidth, introducing a Node-Fusion Intercon-

tion Value (NFIV) to co-locate highly interconnected virtual nodes, reducing virtual links requiring physical provisioning and minimizing embedding cost. Chowdhury *et al.* [3] coordinated node and link mapping so placement decisions directly consider link feasibility and bandwidth cost, enabling more balanced utilization and higher acceptance ratios. Both studies are confined to abstract VNE models excluding optical-layer factors such as spectrum continuity, modulation reach, and IC-XT. To the best of our knowledge, such co-location principles have not been incorporated into optical or SDM-EON slice mapping.

Our work bridges this gap by introducing controlled node co-location within an optical slicing framework that accounts for compute constraints, spectrum allocation, and physical-layer impairments under dynamic traffic. While two baselines (Sorted and Greedy) follow the one-to-one assumption for fair comparison, DPSM and WMSM can be extended to permit controlled co-location by relaxing node-mapping cardinality constraints and augmenting compute-capacity checks.

TABLE I
COMPARISON OF RELATED WORKS BY SDM-EON, COORDINATION, AND CO-LOCATION CHARACTERISTICS

Reference(s)	SDM-EON	Coord.	Co-loc.	Solution Type(s)
[8]–[14]	✓	✗	✗	ILP, Heuristic
[3], [7]	✗	✗	✓	RL, Heuristic
[2]	✗	✓	✓	Heuristic
[4], [5]	✗	✗	✗	ILP, Heuristic
WMSM	✓	✓	✓	Heuristic

III. SYSTEM MODEL AND PROBLEM FORMULATION

We consider an SDM-EON modeled as a graph $G = (V, E)$, where V is the set of physical nodes and E is the set of optical fiber links. Each link supports multiple spatial cores Φ_e , each offering a frequency grid of f discrete spectrum slots. A subset $V_c \subseteq V$ denotes compute-capable nodes with finite processing capacities. Cores are grouped by spatial adjacency to mitigate inter-core crosstalk.

A. Definitions and Operational Semantics

Compute Resource Model. Compute denotes abstract processing capacity at compute-capable nodes, expressed in normalized units representing VNF execution (e.g., transcoding, caching, monitoring). Each slice request specifies an aggregate compute demand without modeling individual VNF internals.

OEO Semantics. Long-haul optical signals accumulate noise and dispersion, eventually degrading below the quality-of-transmission threshold for the chosen modulation format. OEO conversion at a compute-capable waypoint regenerates the signal electrically before re-launching it as a new lightpath, restoring signal integrity and extending transmission reach.

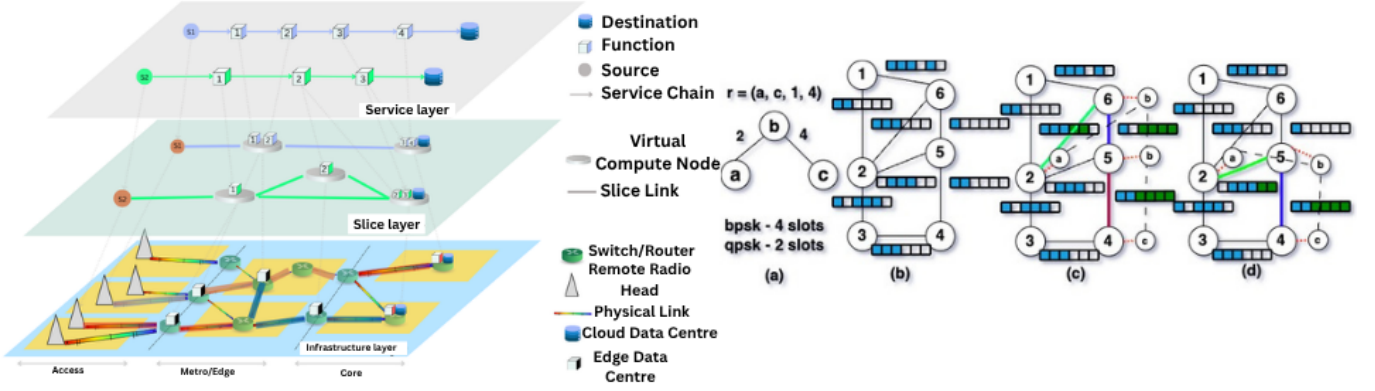


Fig. 1. E2E network slicing and lightpath provisioning.

Deploying a VNF at such a waypoint naturally triggers this conversion. Modulation format selection is a *per-lightpath, connection-time* decision: each segment's transceiver is independently configured at provisioning time based on segment length, so no mid-transmission switching occurs, that is, segments simply use different, statically assigned transceivers. Consequently, spectrum continuity is enforced only within each segment, enabling independent modulation and spectrum assignment per segment and improving feasibility under reach and fragmentation constraints.

Each slice request r is represented as (s, d, b_{req}, c_{req}) , where s and d are source and destination, b_{req} is bandwidth (Gbps), and c_{req} is the compute requirement. A request is accepted if compute and spectrum resources are jointly provisioned along a feasible path or via waypoints, with each allocation decision influencing subsequent ones.

B. Decision Variables

The following integer-valued decision variables govern the slice mapping:

- c_{v,r_i} : a non-negative integer representing the compute units allocated to request r_i at node $v \in V_c$.
- γ_{e,r_i} : a non-negative integer representing the number of spectrum slots allocated to request r_i on link $e \in E$.
- $g_{r_i,k} \in \{0, 1\}$: equals 1 if core group $\mathcal{G}_k$ is assigned to request r_i , and 0 otherwise.
- $M_{r_i}^k \in \{\text{BPSK}, \text{QPSK}, \text{16-QAM}\}$: modulation format selected for segment k of request r_i .

C. Cost Model

For each slice request r_i , the provisioning cost is:

$$C_{r_i} = \sum_{v \in V_c} c_{v,r_i} \mathbb{C}_{compute} + \sum_{e \in E} \gamma_{e,r_i} \mathbb{C}_{spectrum}, \quad (1)$$

where $\mathbb{C}_{compute}$ and $\mathbb{C}_{spectrum}$ are the per-unit costs of compute and spectrum resources, c_{v,r_i} is the integer number

of compute units allocated to r_i at node v , and γ_{e,r_i} is the integer number of spectrum slots allocated to r_i on link e . The overall objective is to minimize the total provisioning cost across all requests:

$$\text{Minimize: } \sum_{i=1}^N C_{r_i}. \quad (2)$$

D. Constraints

The slice mapping must satisfy the following constraints:

- **(C1) Modulation reach:** For each selected path p with $x_{r_i,p}^k = 1$, the total segment length must not exceed the reach of the assigned modulation format:

$$\sum_{e \in p} x_{r_i,p}^k \cdot D_e \leq D_{\max}(m_{r_i}^k), \quad \forall r_i, \forall k \quad (3)$$

where D_e is the length of link e and $D_{\max}(m_{r_i}^k)$ is the maximum transmission reach for modulation format $m_{r_i}^k$.

- **(C2) Spectrum contiguity:** The γ_{e,r_i} slots allocated on each link e must form a contiguous block starting at slot s_{r_i} :

$$\gamma_{e,r_i} = f_{r_i} \cdot x_{r_i,p}^k, \quad \forall e \in p, x_{r_i,p}^k = 1 \quad (4)$$

where the required slot count is:

$$f_{r_i} = b_{req} \times \eta(m_{r_i}^k) \quad (5)$$

with $\eta(m_{r_i}^k)$ being the slots-per-Gbps for modulation format $m_{r_i}^k$, and the allocated slots occupy positions $\{s_{r_i}, s_{r_i} + 1, \dots, s_{r_i} + f_{r_i} - 1\}$.

- **(C3) Spectrum continuity:** The same contiguous slot block must be assigned on every link along a given segment, i.e.,

$$\gamma_{e,r_i} = \gamma_{e',r_i}, \quad \forall e, e' \in p, x_{r_i,p}^k = 1 \quad (6)$$

- **(C4) Compute feasibility:** The allocated compute resources must satisfy

$$\sum_{v \in V_c} c_{v,r_i} \geq c_{req}, \quad \forall r_i, \quad |V_c| \geq 1 \quad (7)$$

$$\sum_{r_i} c_{v,r_i} \leq C_v^{\text{comp}}, \quad \forall v \in V_c \quad (8)$$

where C_v^{comp} is the compute capacity of node v .

- **(C5) Core-group selection (crosstalk avoidance):** Each request is assigned a core $\phi_{r_i} \in \mathcal{G}_k$, where $\mathcal{G}_k$ denotes a non-adjacent core group (e.g., $\mathcal{G}_1 = \{1, 3, 5\}$, $\mathcal{G}_2 = \{2, 4, 6\}$, $\mathcal{G}_3 = \{7\}$), ensuring $\sum_k g_{r_i,k} = 1$ with $g_{r_i,k} = 1$ if $\mathcal{G}_k$ is used by r_i and 0 otherwise. This avoids assigning adjacent cores simultaneously, minimizing inter-core crosstalk.

E. Performance Metrics

The following metrics are used to evaluate performance:

- *Blocking ratio:* fraction of slice requests rejected.
- *Spectrum utilization:*

$$U = \frac{\sum_{i=1}^N \sum_{e \in E} \gamma_{e,r_i}}{\mathbb{N}}, \quad (9)$$

where $\mathbb{N}$ is the total number of slots in the network (links $\times$ cores $\times$ slots per core).

- *Provisioning cost:* average resource cost across accepted requests.

F. Modeling Assumptions

The slice-embedding problem combining RMCSA with compute placement is NP-hard, generalizing RSA [1], VNE [15], and multidimensional bin-packing under modulation reach, crosstalk, and compute capacity constraints [16]. Exact optimization is therefore intractable, motivating the sequential heuristic framework proposed here. Requests may consolidate or distribute virtual functions across nodes [17]; although the framework supports multiple waypoints, this study restricts to at most two for computational tractability.

IV. PROPOSED SOLUTION

We propose a sequential yet integrated heuristic framework for compute-aware slice mapping in SDM-EONs. The overall framework operates in the following ordered stages for each arriving slice request:

- 1) **Path enumeration:** compute k shortest candidate paths between source s and destination d .
- 2) **Compute placement:** identify compute-capable waypoint node(s) along each candidate path satisfying c_{req} ; this partitions the end-to-end path into segments.

- 3) **Modulation selection:** assign a modulation format per segment based on segment length, independently for each segment due to OEO conversion at waypoints.
- 4) **Spectrum allocation:** allocate a contiguous block of γ_{e,r_i} slots on a selected core group per segment, enforcing continuity and contiguity within each segment.
- 5) **Feasibility check:** if any stage fails, backtrack to the next candidate path or waypoint configuration; block the request if all options are exhausted.

Decisions at each stage directly constrain subsequent ones: compute placement determines segment lengths, dictating modulation format, slot count, and spectrum feasibility. This coupling distinguishes the proposed framework from prior decoupled RMCSA approaches. Two strategies instantiate this framework: a baseline DPSM and the proposed WMSM, differing in waypoint selection and segment optimization.

A. Direct Path-Driven Slice Mapping (DPSM)

DPSM computes $k = 3$ shortest paths using Dijkstra's algorithm [11]. For each candidate path p_i , compute feasibility is checked via $\sum_{v \in p_i \cap V_c} c_v \geq c_{req}$; if satisfied, compute resources are mapped onto the first node along p_i meeting c_{req} . Modulation is selected based on total path length D_{p_i} : 16-QAM for $D_{p_i} \leq 500$ km, QPSK for $D_{p_i} \leq 1000$ km, and BPSK otherwise. The required slot count $S = BW \times \eta(M)$ is computed, a core group $\{\mathcal{G}_k\}$ is chosen by the highest available slot ratio, and a contiguous block of S slots is allocated on the first eligible core. If allocation fails, the next path is tried; if all k paths fail, the request is blocked. Among feasible paths, the shortest is selected. DPSM's end-to-end constraint forces a single modulation format, often requiring BPSK and consuming more slots, while its greedy compute placement risks hotspots under high load, making it a conservative benchmark for WMSM.

1) *DPSM with Balanced Placement (DPSM-B):* DPSM-B replaces the greedy compute placement with a segment-length balancing strategy. Since OEO conversion at the compute node creates two independent segments (s, v^*) and (v^*, d) with independently assigned modulation formats, placing the compute node near the path midpoint maximizes the likelihood that both segments qualify for higher-order modulation. For each compute-capable node v along p_i , the segment imbalance is:

$$\delta_v = |d(s, v) - d(v, d)|, \quad (10)$$

and $v^* = \arg \min_v \delta_v$ is selected. When a single node cannot satisfy C_{req} , the algorithm selects the pair (v_1, v_2) minimizing the maximum length among segments (s, v_1) , (v_1, v_2) , and (v_2, d) .

Algorithm 1 Direct Path-Driven Slice Mapping with Balanced placement (DPSM-B)

Input: Slice Request (s, d, BW, C_{req})
Find $k = 3$ shortest paths from s to d using Dijkstra's algorithm
 $feasible \leftarrow \text{false}$
for each path p_i in order **do**
 Calculate total compute capacity along p_i
 if capacity $\geq C_{req}$ **then**
 for each candidate compute node v along p_i **do**
 Compute $\delta_v = |d(s, v) - d(v, d)|$
 end for
 Select node(s) v^* minimizing δ_v
 if multiple nodes needed **then**
 Select combination minimizing max segment length disparity
 end if
 Map compute resources onto v^* to satisfy C_{req}
 // Spectrum checked independently per segment
 Select modulation format for segment (s, v^*) based on $D_1 = d(s, v^*)$
 Compute required slots $S_1 = BW \times (\text{slots per Gbps})$ for (s, v^*)
 Select core group for (s, v^*) with highest available slot ratio
 for each core ϕ_1 in selected group **do**
 if ϕ_1 has S_1 contiguous free slots **then**
 Allocate slots on ϕ_1
 break
 end if
 end for
 Select modulation format for segment (v^*, d) based on $D_2 = d(v^*, d)$
 Compute required slots $S_2 = BW \times (\text{slots per Gbps})$ for (v^*, d)
 Select core group for (v^*, d) with highest available slot ratio
 for each core ϕ_2 in selected group **do**
 if ϕ_2 has S_2 contiguous free slots **then**
 Allocate slots on ϕ_2
 break
 end if
 end for
 if both segments allocated successfully **then**
 $feasible \leftarrow \text{true}$
 break
 end if
 end if
end for
if $feasible = \text{false}$ **then**
 Fallback: execute standard DPSM
return
end if
Update metrics: blocking rate, utilization, cost

Spectrum allocation then proceeds independently per segment: modulation M_k and slot count $S_k = BW \times \eta(M_k)$ are determined from segment length D_k , a core group is selected by the highest available slot ratio, and contiguous slots are allocated on the first eligible core. If either segment fails, the algorithm advances to path p_{i+1} ; if all k paths are exhausted, it falls back to standard DPSM.

B. Waypoint-Assisted Multi-Segment Slice Mapping (WMSM)

WMSM introduces intermediate compute-capable waypoints that decompose each slice request into shorter optical segments, relaxing spectrum continuity to within-segment only and enabling independent modulation selection per segment.

Given a request (s, d, BW, c_{req}) , WMSM enumerates candidate configurations $\mathcal{W} \subseteq V_c$ satisfying $\sum_{w_i \in \mathcal{W}} c_{w_i} \geq c_{req}$, restricted to at most two waypoints. For each configuration, up to $k = 2$ shortest paths are computed per segment (w_i, w_{i+1}) , modulation assigned per segment length D_i (16-QAM: ≤ 500 km, QPSK: ≤ 1000 km, BPSK: otherwise), and slot count $S_i = BW \times \eta(M_i)$ derived accordingly. The configuration minimizing $\sum_i S_i$ is selected as $\mathcal{W}^*$. Spectrum allocation proceeds segment by segment: a core group is chosen by the highest available slot ratio and the first contiguous block of S_i free slots allocated. If any segment fails, the configuration is discarded and the next evaluated; if all fail, the request is blocked. Upon successful allocation, compute capacity is distributed across selected waypoint nodes, with waypoint choice determining segment lengths, modulation, and slot requirements, achieving compute-spectrum coordination without exhaustive joint optimization.

1) *WMSM with Balanced Placement (WMSM-B)*: WMSM-B augments waypoint selection with a segment-length balance objective, since unbalanced segments can force lower-order modulation on longer segments, undermining spectral efficiency. For each candidate configuration $\mathcal{W}$, the imbalance is:

$$\delta_{\mathcal{W}} = \max_i d(w_i, w_{i+1}) - \min_i d(w_i, w_{i+1}), \quad (11)$$

and the selection objective becomes:

$$\mathcal{W}^* = \arg \min_{\mathcal{W}} \left(\sum_{i=0}^{|\mathcal{W}|} d(w_i, w_{i+1}) + \lambda \cdot \delta_{\mathcal{W}} \right), \quad (12)$$

where $\lambda > 0$ trades off total path length against segment balance. Per-segment modulation, slot allocation, and fallback proceed identically to baseline WMSM. If allocation fails, the algorithm backtracks through configurations ranked by the augmented objective until a feasible mapping is found or all options are exhausted.

Algorithm 2 Waypoint-Assisted Multi-Segment Slice Mapping with Balanced PLacement (WMSM-B)

Input: Slice request (s, d, BW, C_{req})

Output: Mapping decision (accepted/rejected), spectrum and compute allocation

Identify a candidate set of compute-capable waypoints $\mathcal{W}$ such that total available capacity $\sum_{w_i \in \mathcal{W}} C_{w_i} \geq C_{req}$
 For each candidate waypoint configuration, compute segment balance:

$$\delta_{\mathcal{W}} = \max_i d(w_i, w_{i+1}) - \min_i d(w_i, w_{i+1}),$$

where $w_0 = s, w_{|\mathcal{W}|+1} = d$

Select waypoint(s) $\mathcal{W}^*$ that jointly minimize total end-to-end distance and segment imbalance:

$$\min_{\mathcal{W}} \left(\sum_{i=0}^{|\mathcal{W}|} d(w_i, w_{i+1}) + \lambda \cdot \delta_{\mathcal{W}} \right),$$

where $w_0 = s, w_{|\mathcal{W}|+1} = d$

▷ For simplicity, we consider up to two waypoints in this paper

for each segment between consecutive nodes (w_i, w_{i+1}) **do**

Compute up to $k = 2$ shortest paths

Estimate distance D_i for each path

Select feasible modulation format based on D_i

Compute required slots $S_i = BW \times (\text{slots per Gbps})$

end for

Combine feasible segment paths to form candidate routes

Choose the route minimizing total spectrum usage $\sum_i S_i$

for each segment in the selected route **do**

for each core ϕ in the assigned core group **do**

if ϕ has S_i contiguous free slots **then**

Allocate S_i slots on ϕ

break

end if

end for

if no allocation is feasible **then**

Fallback: execute standard WMSM

return

end if

end for

for each selected waypoint $w_i \in \mathcal{W}^*$ **do**

Allocate compute capacity $C_{req,i}$ ensuring $\sum_i C_{req,i} = C_{req}$

end for

Update network metrics: blocking rate, utilization, cost

Both WMSM and WMSM-B enforce spectrum continuity, contiguity, and non-overlap within each segment, with inter-core crosstalk mitigated via group-based core selection. For example, a 1500 km path requiring QPSK end-to-end can be

split into 700 km and 800 km segments, both feasible under 16-QAM, halving the spectrum cost. WMSM's complexity scales polynomially with network size and linearly with candidate waypoints, making it practical for dynamic SDM-EONs.

C. Baseline Approaches

1) *Sorted Approach:* Inspired by the strategy proposed in [10], we implement a baseline approach: the request prioritization mechanism, where slice requests are first sorted based on their compute and bandwidth demands. Prior to resource allocation, physical nodes are ranked by available compute capacity, while adjacent links are ordered according to spectrum slot availability. Unlike [10], which considers virtual topology cycles and maps them onto matching physical cycles, our implementation simplifies the scenario by treating each request as an aggregate compute demand without requiring cycle-aware embedding.

2) *Greedy Approach:* A greedy baseline approach is also implemented, where slice requests are provisioned using a first-come, first-served (FCFS) strategy. For compute resource allocation, the first physical node encountered along the candidate path that satisfies the compute requirement is selected. Similarly, for spectrum allocation, the first available core on each traversed link that can accommodate the required contiguous spectrum slots is chosen, without global optimization across requests.

V. NUMERICAL EVALUATION

A. Simulation Setup

Table II summarizes the simulation parameters. Bandwidth demands of 1–20 Gbps and compute needs of 5–10 units represent light to moderate workloads (transcoding, caching, monitoring) [18], corresponding to 1–20 spectrum slots depending on modulation [11]. Simulations use the 14-node NSFNET topology [9] with 7 cores and 120 slots per link, distances of 350–1500 km, and three modulation formats: BPSK (2000 km, 4 slots/Gbps), QPSK (1000 km, 2 slots/Gbps), and 16-QAM (500 km, 1 slot/Gbps). Spectrum and compute unit costs are 40 and 400, respectively. Compute capacities are set high (4000 units/node) to isolate spectrum effects, then reduced (400 units/node) to evaluate coupled compute–spectrum constraints, with 11 randomly selected compute-capable nodes.

Requests arrive as a Poisson process with exponentially distributed holding times (mean 0.5 hrs), with 50,000 requests per experiment between random source–destination pairs. The 1–20 Gbps range ensures feasibility under BPSK (20 Gbps occupies 80 slots, within the 120-slot limit); experiments up to 40 Gbps showed consistent trends but are omitted due to page constraints. Routing uses $k = 3$ shortest paths

TABLE II
SIMULATION PARAMETERS

Parameter	Value
Topology	NSFNET (14 nodes, 21 links)
Cores per link	7
Spectrum slots per core	120
Link distances	350–1500 km
Modulation: BPSK	reach 2000 km, 4 slots/Gbps
Modulation: QPSK	reach 1000 km, 2 slots/Gbps
Modulation: 16-QAM	reach 500 km, 1 slot/Gbps
Bandwidth demand b_{req}	1–20 Gbps (uniform)
Compute demand c_{req}	5–10 units (uniform)
Compute capacity (high)	4000 units/node
Compute capacity (limited)	400 units/node
Compute-capable nodes	11 (randomly selected)
Spectrum slot cost	40 per Gbps
Compute resource cost	400 per unit
Request arrival	Poisson process
Holding time	Exponential (mean 0.5 hrs)
Requests per experiment	50,000
Independent runs	30
Observed variance	<5%
Shortest paths k (DPSM)	3
Shortest paths k (WMSM)	2 per segment

in DPSM and $k = 2$ per segment in WMSM. Each data point is averaged over 30 runs with variance below 5%. Prior RMCSA works [12] confirm that algorithms similar to WMSM reduce fragmentation and improve fairness, consistent with our observations.

We compare our approaches against a prioritization strategy inspired by [10], where requests are sorted by compute and bandwidth requirements, nodes ranked by compute capacity, and links by spectrum availability. Unlike cycle-aware embedding in [10], our baseline treats requests as aggregate compute demands, providing a conservative benchmark for DPSM and WMSM.

While global optimality is not guaranteed, near-optimal behavior is supported on two grounds: sequential compute–spectrum coupling ensures each decision accounts for downstream feasibility, and the substantial performance gap over Sorted/Greedy baselines suggests the heuristic operates well within the feasible region. An ILP formulation is possible for small static instances but becomes intractable at the scale evaluated here (50,000 dynamic requests, 14-node 7-core topology). Sorted and Greedy also serve as an implicit ablation: by decoupling compute and spectrum decisions, they approximate a pure RMCSA baseline, and the performance gap provides evidence of the benefit of joint optimization; a full ablation is deferred to a journal extension.

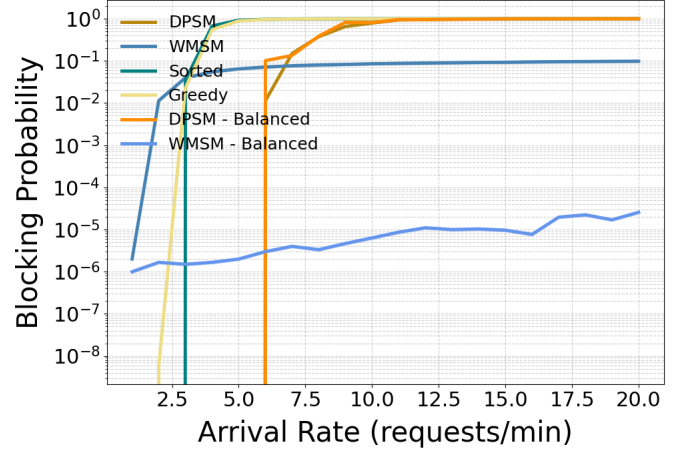


Fig. 2. Blocking Rate vs. Arrival Rate.

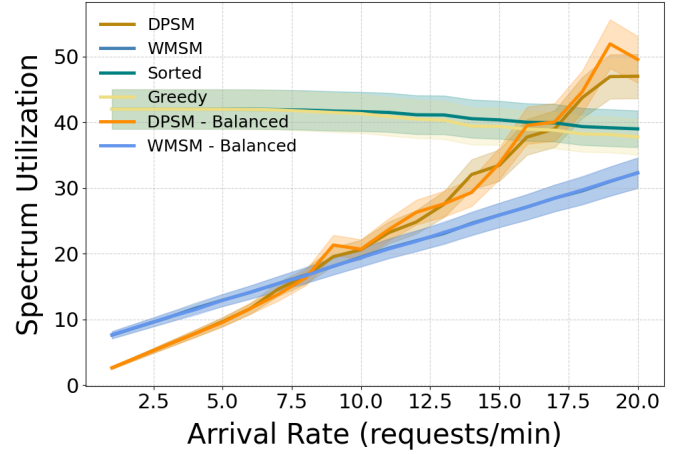


Fig. 3. Spectrum Utilization vs. Arrival Rate.

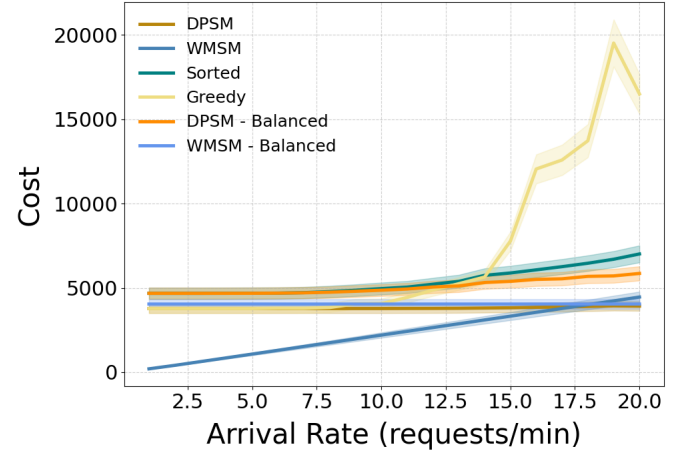


Fig. 4. Provisioning Cost vs. Arrival Rate.

B. Results and Discussion

Figures 2, 3, and 4 show blocking rate, spectrum utilization, and cost under high compute availability (4000 units per

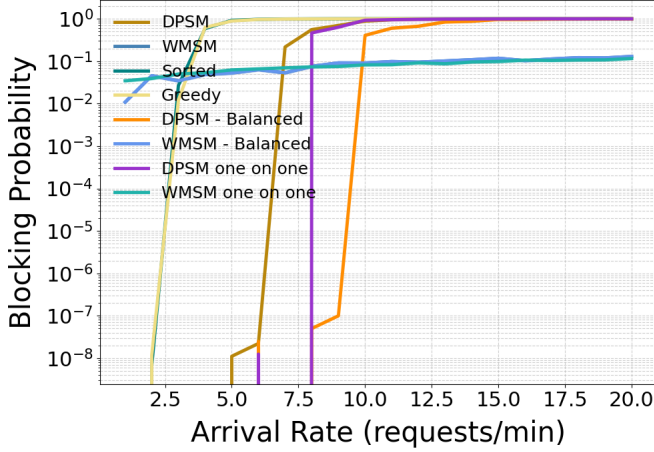


Fig. 5. Blocking Rate vs. Arrival Rate for limited compute capacity.

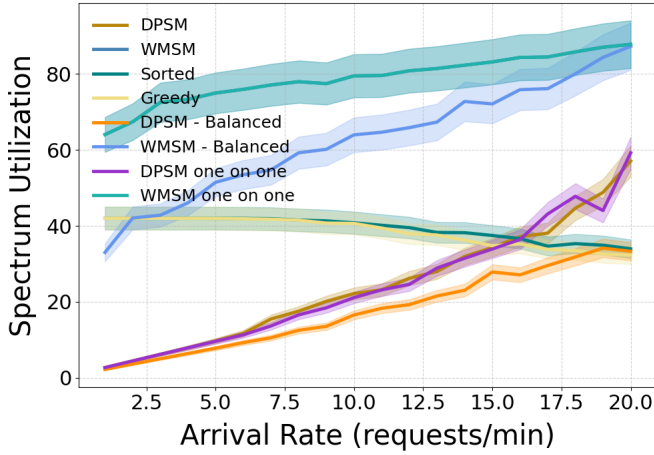


Fig. 6. Spectrum Utilization vs. Arrival Rate for limited compute capacity.

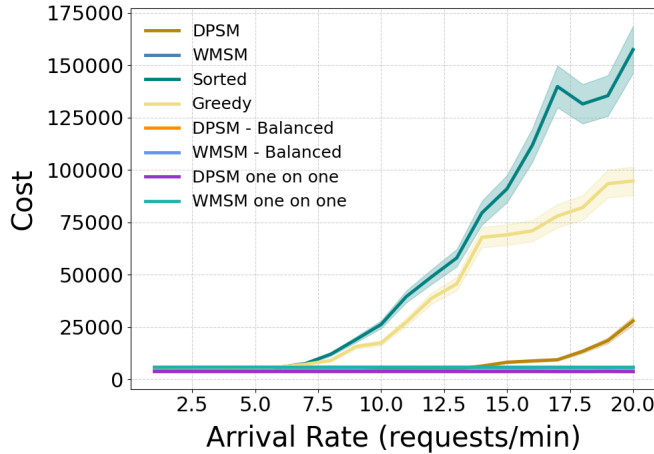


Fig. 7. Provisioning Cost vs. Arrival Rate for limited compute capacity.

node). Figures 5, 6, and 7 present the same metrics when

compute capacity is limited to 400 units per node.

Approaches compared:

- **One-on-one:** Maps each virtual node to a distinct physical node without fusion, serving as a baseline to isolate the benefit of node fusion, where merging multiple virtual nodes onto a single physical node reduces resource consumption and improves scalability.
- **With Balanced Placement:** Integrates VONE and RM-CSA sequentially with a balanced placement mechanism, where spectrum allocation outcomes are informed with refined node mapping decisions, unlike conventional decoupled approaches that solve the two stages independently.
- **Sorted and Greedy:** Standard baseline heuristics for RMCSA that assign spectrum slots in sorted or greedy order without any awareness of virtual topology or waypoint structure, included to benchmark the gain from our proposed decomposition strategies.

The results highlight three key insights:

- **Blocking (Figs. 2, 5):** In Fig. 2, WMSM with Balanced Placement achieves the lowest blocking (10^{-8}) across all arrival rates. Standard WMSM sustains 10^{-6} – 10^{-5} , while DPSM with Balanced Placement remains near-zero until ~ 6 req/min before rising sharply to 10^{-1} . Standard DPSM stabilizes at 10^{-1} from early rates; Sorted and Greedy saturate to 10^0 at ~ 3 req/min. In Fig. 5, under limited capacity, WMSM one-on-one and WMSM with Balanced Placement sustain the lowest blocking (10^{-5}), while DPSM variants saturate to 10^0 around 8–9 req/min and Sorted/Greedy collapse at ~ 3 req/min, confirming capacity constraints disproportionately impact less adaptive schemes.
- **Spectrum utilization (Figs. 3, 6):** In Fig. 3, Sorted and Greedy maintain high but declining utilization ($\sim 40\%$) as they saturate early. DPSM and DPSM with Balanced Placement grow steadily to $\sim 47\%$ and $\sim 51\%$ at 20 req/min, with the latter exhibiting higher variance at peak loads. WMSM and WMSM with Balanced Placement achieve the lowest utilization, growing from $\sim 8\%$ to $\sim 32\%$, confirming superior spectrum efficiency across all traffic intensities; WMSM and WMSM one-on-one track closely, indicating convergent resource consumption. In Fig. 6, Sorted and Greedy plateau near 42% before declining as blocking increases. DPSM and DPSM one-on-one grow aggressively to $\sim 57\%$ at peak loads with high variance, while WMSM and WMSM one-on-one remain the most efficient ($\sim 8\%$ – 32%), confirming multi-segment decomposition retains its advantage under constrained resources.
- **Provisioning cost (Figs. 4, 7):** In Fig. 4, WMSM with

Balanced Placement achieves the flattest cost curve while Sorted and Greedy escalate rapidly and DPSM rises moderately; WMSM-based schemes maintain near-flat profiles across all arrival rates. In Fig. 7, disparities become more pronounced: Sorted and Greedy exceed $\sim 150,000$ at 20 req/min, DPSM reaches $\sim 27,000$, while WMSM and WMSM with Balanced Placement maintain near-flat profiles ($\sim 4,000$), confirming multi-segment decomposition provides the most cost-stable provisioning under constrained capacity.

Although fragmentation and fairness results are not plotted, WMSM exhibits lower fragmentation than DPSM by balancing allocations across cores, consistent with group-processing RMCSA studies emphasizing fairness improvements from prioritization.

While evaluated on NSFNET, WMSM's localized decision-making, that is, shortest-path computation, waypoint selection, and segment-level feasibility checks, is not tailored to any topology-specific property, and its benefits are expected to generalize to topologies such as USNET or COST239 with different diameters, node degrees, and link lengths.

In summary, DPSM provides lightweight efficiency in balanced scenarios while WMSM achieves superior performance under high loads by trading additional transceivers for spectrum savings, with both approaches outperforming the baselines and demonstrating the benefit of compute-aware slice mapping in SDM-EONs.

VI. CONCLUSION

This paper presented DPSM and WMSM for compute-aware SDM-EON provisioning. WMSM consistently outperforms DPSM and Sorted/Greedy baselines across blocking probability, spectrum utilization, and provisioning cost, owing to segment-by-segment optimization and dynamic compute placement. Balanced placement further improves performance, sustaining blocking as low as 10^{-5} under limited capacity, while node fusion yields meaningful reductions in resource consumption and cost. Under unlimited capacity, WMSM maintains near-flat cost profiles while Sorted and Greedy escalate rapidly; under limited capacity, these disparities become more pronounced, validating robustness under constrained resources. Future work includes composite compute function mapping, integrating compute and spectrum availability metrics into provisioning decisions, explicit fragmentation optimization, and Reinforcement Learning-based optimal solutions.

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